SUPPORT OF A RELATIVE CYCLE ALONG A SMOOTH MORPHISM OF RELATIVE DIMENSION R IS OPEN AND CLOSED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Definition of base change of cycles over fields, under which the coefficients of components of a reduced base change are read off from the original cycle (relative-cycles chapter, Section section-relative-fields, canonical tag 0H4E). cited at proof lines 1274-1275. Yields the invariance of the coefficient n_x under base change: the smooth component Z(x) pulls back to a reduced scheme and Z(x') is an irreducible component of it.
- Specializations in a locally Noetherian scheme can be realized by a morphism from the spectrum of a discrete valuation ring (Properties chapter, Lemma properties-lemma-locally-Noetherian-specialization-dvr, canonical tag 054F). cited at proof lines 1282-1285. Provides the morphism S' to X mapping the generic point to x' and the closed point to x, turning the specialization check into a valuation-ring computation.
- Connected components of fibres along a section (More on Morphisms, Section more-morphisms-section-connected-components, canonical tag 055K, and especially Lemma more-morphisms-lemma-connected-along-section-open, canonical tag 055R). cited at proof lines 1295-1298. After replacing X by an open subscheme the fibres of X to S are connected, using the section sigma of the base-changed family.

Source reference. Original section 0H4E: sources/stacks-chow/tags.txt, source lines 21349–21349. Cited at proof lines 1274-1275: the coefficient invariance under base change rests on the definition of base change given there.

Source reference. Original lemma 054F: sources/stacks-chow/tags.txt, source lines 6638–6638. Cited at proof lines 1282-1285: realizes the specialization x' leading to x by a valuation ring.

Source reference. Original section 055K: sources/stacks-chow/tags.txt, source lines 6678–6678. Cited at proof lines 1295-1296: the discussion used to shrink X so its fibres are connected.

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 055R: sources/stacks-chow/tags.txt, source lines 6684–6684. Cited at proof lines 1297-1298: provides the open subscheme with connected fibres in the DVR case.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Let $r \geq 0$. Assume S locally Noetherian and f smooth of relative dimension r. Let $\alpha \in z(X/S, r)$. Then the support of α is open and closed in X (see proof for a more precise result).*

Proof. Let $x \in X$ with image $s \in S$. Since f is smooth, there is a unique irreducible component $Z(x)$ of X_s which contains x. Then $\dim(Z(x)) = r$. Let n_x be the coefficient of $Z(x)$ in the cycle α_s. We will show the function $x \mapsto n_x$ is locally constant on X.

Let $g : S' \to S$ be a morphism of locally Noetherian schemes. Let X' be the base change of X and let $\alpha' = g^*\alpha$ be the base change of α. Let $x' \in X'$ map to $s' \in S'$, $x \in X$, and $s \in S$. We claim $n_{x'} = n_x$. Namely, since $Z(x)$ is smooth over $\kappa(s)$ we see that $Z(x) \times_{\mathrm{Spec}(\kappa(s))} \mathrm{Spec}(\kappa(s'))$ is reduced. Since $Z(x')$ is an irreducible component of this scheme, we see that the coefficient $n_{x'}$ of $Z(x')$ in $\alpha'_{s'}$ is the same as the coefficient n_x of $Z(x)$ in α_s by the definition of base change in Section 0H4E thereby proving the claim.

Since X is locally Noetherian, to show that $x \mapsto n_x$ is locally constant, it suffices to show: if $x' \rightsquigarrow x$ is a specialization in X, then $n_{x'} = n_x$. Choose a morphism $S' \to X$ where S' is the spectrum of a discrete valuation ring mapping the generic point η to x' and the closed point 0 to x. See Properties, Lemma 054F. Then the base change $X' \to S'$ of f by $S' \to S$ has a section $\sigma : S' \to X'$ such that $\sigma(\eta) \rightsquigarrow \sigma(0)$ is a specialization of points of X' mapping to $x' \rightsquigarrow x$ in X. Thus we reduce to the claim in the next paragraph.

Let S be the spectrum of a discrete valuation ring with generic point η and closed point 0 and we have a section $\sigma : S \to X$. Claim: $n_{\sigma(\eta)} = n_{\sigma(0)}$. By the discussion in More on Morphisms, Section 055K and especially More on Morphisms, Lemma 055R after replacing X by an open subscheme, we may assume the fibres of $X \to S$ are connected. Since these fibres are smooth, they are irreducible. Then we see that $\alpha_\eta = n[X_\eta]$ with $n = n_{\sigma(\eta)}$ and the relation $sp_{X/S}(\alpha_\eta) = \alpha_0$ implies $\alpha_0 = n[X_0]$, i.e., $n_{\sigma(0)} = n$ as desired. $\square$